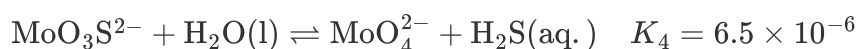
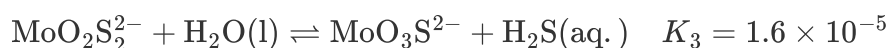
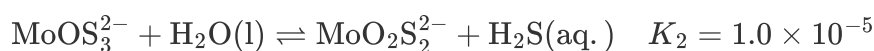
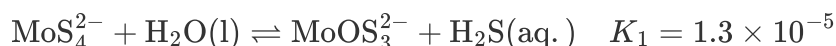


Question

The following reactions control the stepwise hydrolysis and sulfur substitution sequence of molybdate and thiomolybdate in dilute aqueous solution (the concentration of water is already included in the equilibrium constant):



MoS_4^{2-} with an initial concentration of $2.00 \times 10^{-7} \text{ mol L}^{-1}$ undergoes hydrolysis in a buffer solution at $\text{pH} = 5.00$. Considering the ionization and volatilization of $\text{H}_2\text{S}(\text{aq.})$, it is known that:

- The buffer capacity is large enough that the pH does not change, and the buffer species do not interfere with the given reactions.
- $\text{p}K_{a1} = 6.88$, $\text{p}K_{a2} = 12.90$ for H_2S
- The solution volume is $V_{\text{aq}} = 1.00 \text{ L}$, and the gas phase volume above is $V_{\text{g}} = 5.00 \text{ L}$ (the whole system is closed)
- Henry's constant for H_2S is $H = 9.741 \times 10^{-3} \text{ atm}/(\text{mol} \cdot \text{m}^{-3})$
- Take $T = 298.15 \text{ K}$, and the ideal gas constant is $R = 0.082057 \text{ L atm mol}^{-1} \text{ K}^{-1}$

Based on the above information, calculate the concentration of each species, and based on the concentration of each species, calculate the ratio of the product of the concentrations/partial pressures of all molybdenum-containing species to the product of the concentrations of all sulfur-containing, molybdenum-free species (normalization is required, the concentration is in $\frac{c}{c^\ominus}$, and the partial pressure is in $\frac{p}{p^\ominus}$). The final result should retain the appropriate number of significant figures, and choose the option within 10% of the calculated result; otherwise, choose option A: All other options are incorrect.

A. All other options are incorrect

B. 3.1×10^{-5}

C. 7.7×10^{-8}

D. 2.2×10^{-10}

E. 3.9×10^{-12}

F. 4.3×10^{-13}

G. 9.1×10^{-14}

H. 3.1×10^{-16}

I. 5.9×10^{-18}

J. 9.3×10^{-20}

K. 1.7×10^{-25}

L. 2.8×10^{-28}

Answer

Correct Answer: **F**

Detailed Explanation

Known

- Initial: $[\text{MoS}_4^{2-}]_0 = C_0 = 2.00 \times 10^{-7} \text{ mol L}^{-1}$
- Equilibrium constants for stepwise hydrolysis of MoS_4^{2-} :
 $K_1 = 1.3 \times 10^{-5}$, $K_2 = 1.0 \times 10^{-5}$, $K_3 = 1.6 \times 10^{-5}$, $K_4 = 6.5 \times 10^{-6}$
- The pH of the solution is fixed at 5.00
- Ionization of H_2S : $\text{p}K_{a1} = 6.88$, $\text{p}K_{a2} = 12.90$.
- System volume: solution $V_{\text{aq}} = 1.00 \text{ L}$, gas phase $V_g = 5.00 \text{ L}$
- Henry's constant (original units): $H = 9.741 \times 10^{-3} \text{ atm}/(\text{mol m}^{-3})$. Converted to the form with mol L^{-1} yields $H' = H \times 1000 = 9.741 \text{ atm L mol}^{-1}$
- Take $T = 298.15 \text{ K}$, $R = 0.082057 \text{ L atm mol}^{-1} \text{ K}^{-1}$

Let the concentration of molecular H_2S in the solution at equilibrium be $x = [\text{H}_2\text{S}(\text{aq})]$.

CHECKPOINT

0.5 PTS

Let the concentration of molecular H_2S in the solution at equilibrium be $x = [\text{H}_2\text{S}(\text{aq})]$

Define the concentrations of Mo species:

$$[A_0] = [\text{MoS}_4^{2-}], [A_1] = [\text{MoOS}_3^{2-}], [A_2] = [\text{MoO}_2\text{S}_2^{2-}], [A_3] = [\text{MoO}_3\text{S}^{2-}], [A_4] = [\text{MoO}_4^{2-}].$$

Express each Mo species using the equilibrium constant, from the reaction equilibrium (using concentrations instead of activities):

$$[A_1] = \frac{K_1}{x} [A_0]$$

$$[A_2] = \frac{K_1 K_2}{x^2} [A_0]$$

$$[A_3] = \frac{K_1 K_2 K_3}{x^3} [A_0]$$

$$[A_4] = \frac{K_1 K_2 K_3 K_4}{x^4} [A_0]$$

Let:

$$D(x) = 1 + \frac{K_1}{x} + \frac{K_1 K_2}{x^2} + \frac{K_1 K_2 K_3}{x^3} + \frac{K_1 K_2 K_3 K_4}{x^4}$$

Then the molybdenum conservation gives:

$$[A_0] = \frac{C_0}{D(x)}$$

CHECKPOINT

1 PTS

$[\text{MoS}_4^{2-}]$ and the relationship between the total Mo concentration is $[A_0] = \frac{C_0}{D(x)}$

Let:

$$S(x) = \frac{K_1}{x} + \frac{2K_1K_2}{x^2} + \frac{3K_1K_2K_3}{x^3} + \frac{4K_1K_2K_3K_4}{x^4},$$

Then the "total equivalents of H_2S produced" can be written from the equivalents of H_2S generated in each step (calculated per liter of solution):

$$\text{produced } \text{H}_2\text{S} = \frac{C_0}{D(x)} S(x)$$

CHECKPOINT

1 PTS

The relationship for the total equivalents of H_2S produced is $\text{produced } \text{H}_2\text{S} = \frac{C_0}{D(x)} S(x)$

Further, consider the partitioning relationship of H_2S between the solution and the gas phase (under the condition of fixed $\text{pH} = 5$)

In the solution, the total dissolved sulfur equivalents (in H_2S equivalents) is:

$$c_{\text{aq}} = x(1 + \alpha_1 + \alpha_2)$$

Where:

$$\alpha_1 = \frac{[\text{HS}^-]}{[\text{H}_2\text{S}]} = 10^{\text{pH} - \text{p}K_{a1}}$$

$$\alpha_2 = \frac{[\text{S}^{2-}]}{[\text{H}_2\text{S}]} = 10^{2\text{pH} - \text{p}K_{a1} - \text{p}K_{a2}}$$

CHECKPOINT

1 PTS

According to $\text{p}K_a$, the relationship for the total dissolved sulfur equivalents is $c_{\text{aq}} = x(1 + \alpha_1 + \alpha_2)$

The partial pressure of H_2S in the gas phase is given by Henry's law:

$$p = H' x$$

The number of moles in the gas phase (calculated per liter of solution):

$$n_{\text{g per L}} = \frac{p V_g}{RT} = \frac{H' x V_g}{RT}$$

Overall mass conservation (H_2S equivalents):

$$\frac{C_0}{D(x)} S(x) = x(1 + \alpha_1 + \alpha_2) + \frac{H' x V_g}{RT}$$

CHECKPOINT

1 PTS

The overall mass conservation relationship for H_2S is $\frac{C_0}{D(x)} S(x) = x(1 + \alpha_1 + \alpha_2) + \frac{H' x V_g}{RT}$

Simplified, we obtain the equation (with unknown x):

$$x = \frac{C_0 S(x)}{D(x) F}$$

Where:

$$F \equiv \left(1 + \alpha_1 + \alpha_2\right) + \frac{H'V_g}{RT}.$$

CHECKPOINT

1 PTS

Obtain the equation with unknown x , $x = \frac{C_0 S(x)}{D(x) F}$, $F \equiv \left(1 + \alpha_1 + \alpha_2\right) + \frac{H'V_g}{RT}$

Further, use numerical root-finding methods such as the bisection method or Newton's method to solve this equation, $f(x) = \frac{C_0 S(x)}{D(x)} - x \cdot F = 0$.

First, calculate the obtained intermediate constant values:

$$\alpha_1 = 10^{5.00-6.88} = 1.3183 \times 10^{-2}$$

$$\alpha_2 = 10^{10.00-19.78} = 1.6596 \times 10^{-10}$$

$$H' = 9.741 \text{ atm} \cdot \text{L mol}^{-1}$$

$$\frac{H'V_g}{RT} \approx 1.9908$$

$$F \approx (1 + \alpha_1 + \alpha_2) + 1.9908 \approx 3.0040$$

CHECKPOINT

1 PTS

$$F \approx (1 + \alpha_1 + \alpha_2) + 1.9908 \approx 3.0040$$

Finally, solving yields:

$$x = [\text{H}_2\text{S}(\text{aq})] = 2.6363 \times 10^{-7} \text{ mol L}^{-1}$$

CHECKPOINT

3 PTS

The concentration of H_2S is $x = 2.6363 \times 10^{-7} \text{ mol L}^{-1}$

Next, use x to calculate the concentrations of other species, obtaining:

$$D(x) \approx 2.914291 \times 10^6$$

$$[A_0] = [\text{MoS}_4^{2-}] = \frac{C_0}{D(x)} \approx 6.8627 \times 10^{-14} \text{ mol L}^{-1}$$

CHECKPOINT

0.5 PTS

$$[\text{MoS}_4^{2-}] = 6.8627 \times 10^{-14} \text{ mol L}^{-1}$$

$$[A_1] = [\text{MoOS}_3^{2-}] = \frac{K_1}{x} [A_0] \approx 3.3841 \times 10^{-12} \text{ mol L}^{-1}$$

CHECKPOINT

0.5 PTS

$$[\text{MoOS}_3^{2-}] = 3.3841 \times 10^{-12} \text{ mol L}^{-1}$$

$$[A_2] = [\text{MoO}_2\text{S}_2^{2-}] \approx 1.2836 \times 10^{-10} \text{ mol L}^{-1}$$

CHECKPOINT

0.5 PTS

$$[\text{MoO}_2\text{S}_2^{2-}] = 1.2836 \times 10^{-10} \text{ mol L}^{-1}$$

$$[A_3] = [\text{MoO}_3\text{S}^{2-}] \approx 7.7905 \times 10^{-9} \text{ mol L}^{-1}$$

CHECKPOINT

0.5 PTS

$$[\text{MoO}_3\text{S}^{2-}] = 7.7905 \times 10^{-9} \text{ mol L}^{-1}$$

$$[A_4] = [\text{MoO}_4^{2-}] \approx 1.9208 \times 10^{-7} \text{ mol L}^{-1}$$

CHECKPOINT

0.5 PTS

$$[\text{MoO}_4^{2-}] = 1.9208 \times 10^{-7} \text{ mol L}^{-1}$$

Ionization products in solution:

$$[\text{HS}^-] = \alpha_1 x \approx 3.4754 \times 10^{-9} \text{ mol L}^{-1}$$

$$[\text{S}^{2-}] = \alpha_2 x \approx 4.3752 \times 10^{-17} \text{ mol L}^{-1}$$

CHECKPOINT

1 PTS

$$[\text{HS}^-] = \alpha_1 x \approx 3.4754 \times 10^{-9} \text{ mol L}^{-1}$$

CHECKPOINT

1 PTS

$$[\text{S}^{2-}] = \alpha_2 x \approx 4.3752 \times 10^{-17} \text{ mol L}^{-1}$$

Gas phase species:

$$p = H'x \approx 2.5680 \times 10^{-6} \text{ atm}$$

CHECKPOINT

0.5 PTS

 Partial pressure of H_2S gas $p = 2.5680 \times 10^{-6} \text{ atm}$

Verify the correctness of the results by verifying the material balance of Mo:

$$\sum_{i=0}^4 [A_i] = 2.0000 \times 10^{-7} \text{ M} = C_0$$

CHECKPOINT

0.5 PTS

Verify $\sum_{i=0}^4 [A_i] = C_0$

Finally, according to the requirements of the question, calculate:

$$P_S = \frac{[\text{H}_2\text{S}(\text{aq})]}{c^\ominus} \cdot \frac{[\text{HS}^-]}{c^\ominus} \cdot \frac{[\text{S}^{2-}]}{c^\ominus} \cdot \frac{p(\text{H}_2\text{S})}{p^\ominus} \approx 1.02944 \times 10^{-37}$$

$$P_{\text{Mo}} = \prod_{i=0}^4 \frac{[A_i]}{c^\ominus} = \frac{[\text{MoS}_4^{2-}] [\text{MoOS}_3^{2-}] [\text{MoO}_2\text{S}_2^{2-}] [\text{MoO}_3\text{S}^{2-}] [\text{MoO}_4^{2-}]}{(c^\ominus)^5} \approx 4.46088 \times 10^{-50}$$

According to the question, retain two significant figures, and the final ratio is:

$$\frac{P_{\text{Mo}}}{P_S} = 4.3333 \times 10^{-13} \approx 4.3 \times 10^{-13}$$

CHECKPOINT

2 PTS

The final ratio is 4.3×10^{-13}